In general (ignoring the details of the packaging and scheduling, such as a Kubernetes pod versus a Lambda function), cloud-native systems have the following characteristics:

- *They are distributed in nature.* Most of the communication is not in process—and maybe not even on the same machine (e.g., using inter-process communication [IPC] mechanisms)—but takes place via the network.
- *Due to the distributed nature, not only the* what *but also the* where *is crucial.* Think of a number of microservices along the request path serving a specific user. You want to be able to understand what's going on end to end and also be able to drill down into each of the services.
- *The volatility of the services is higher compared to a monolith running on, say, a VM.* Pods (and, with it, their IP addresses) come and go, new versions of a Lambda function might be pushed out every hour, and so on.

Let's now look at how observability can help address the challenges found in and with cloud-native systems. With cloud native systems, you don't want to fly blind; having an automated way to collect all relevant signals and use them as input for decisions—that is, actionable insight—is the ideal.

1.5.1 Return on investment

To understand the right level of investment (i.e., time and money) into observability, you need to know about the costs first:

- *Instrumentation costs*—These include developers' time, which is an ongoing cost.
- *Signal retention*—These costs (e.g., log storage) can be one-off but usually fall under the pay-as-you-go model.
- *Overhead of agents and instrumentation*—This manifests in compute and memory resource usage on top of what the service itself uses. Costs should be in the low single-digit range of CPU and RAM utilization. For example, the AWS OpenTelemetry Collector (http://mng.bz/N2J1) provides a report on the footprint across different signal types.
- *Network usage*—Costs of this type, such as egress traffic, may be a significant cost driver for observability solutions.

The return on the investment into observability is harder to calculate. Determining your return depends mainly on how clear the goals are. We will discuss the goals of different roles shortly. For example, you can measure the mean time to recovery (MTTR; https://www.atlassian.com/incident-management/kpis/common-metrics) for a given service before and after instrumentation. But there are also many human-related factors (e.g., on-call stress or work–life balance) that count toward the gain. You can read more about the topic in "Return on Investment Driven Observability" (https://arxiv.org/abs/2303.13402), a short paper I published on the topic.